

The total number of marks available for this examination paper is 80. It contributes 40% to AS Physics and 20% to the Advanced GCE in Physics.

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**Calculator**

It is recommended that students have access to a scientific calculator for this paper.

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**Formulae sheet**

Students will be provided with the formulae sheet shown in *Appendix 8: Formulae*. Any other physics formulae that are required will be stated in the question paper.

### 8.3 The Sound of Music (MUS)

A study of music and recorded sound, focusing on the production of sound by musical instruments and the operation of a CD player:

- synthesised and 'live' sounds
- standing waves in string and wind instruments
- reading a CD by laser.

Waves and photons are used to model the behaviour of light.

There are opportunities for students to develop ICT skills and other skills relating to practical investigation and to communication.

Students discuss environmental issues related to noise.

| Students will be assessed on their ability to:                                                                           | Suggested experiments                                                               |
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| 28 understand and use the terms <i>amplitude</i> , <i>frequency</i> , <i>period</i> , <i>speed</i> and <i>wavelength</i> | Wave machine or computer simulation of wave properties                              |
| 29 identify the different regions of the electromagnetic spectrum and describe some of their applications                |                                                                                     |
| 30 use the wave equation $v = f\lambda$                                                                                  |                                                                                     |
| 31 recall that a sound wave is a longitudinal wave which can be described in terms of the displacement of molecules      | Demonstration using a loudspeaker<br><br>Demonstration using waves on a long spring |
| 32 use graphs to represent transverse and longitudinal waves, including standing waves                                   |                                                                                     |
| 33 explain and use the concepts of wavefront, coherence, path difference, superposition and phase                        | Demonstration with ripple tank                                                      |
| 34 recognise and use the relationship between phase difference and path difference                                       |                                                                                     |

| Students will be assessed on their ability to:                                                                                                                                                                                                              | Suggested experiments                                     |
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| 35 explain what is meant by a <i>standing (stationary) wave</i> , investigate how such a wave is formed, and identify nodes and antinodes                                                                                                                   | Melde's experiment, sonometer                             |
| 36 recognise and use the expression for refractive index $\mu_2 = \sin i / \sin r = v_1 / v_2$ , determine refractive index for a material in the laboratory, and predict whether total internal reflection will occur at an interface using critical angle |                                                           |
| 37 investigate and explain how to measure refractive index                                                                                                                                                                                                  | Measure the refractive index of both solids and liquids   |
| 38 discuss situations that require the accurate determination of refractive index                                                                                                                                                                           |                                                           |
| 39 investigate and explain what is meant by <i>plane polarised light</i>                                                                                                                                                                                    | Models of structures to investigate stress concentrations |
| 40 investigate and explain how to measure the rotation of the plane of polarisation                                                                                                                                                                         |                                                           |
| 44 recall that, in general, waves are transmitted and reflected at an interface between media                                                                                                                                                               | Demonstration using a laser                               |
| 45 explain how different media affect the transmission/reflection of waves travelling from one medium to another                                                                                                                                            |                                                           |
| 63 explain how the behaviour of light can be described in terms of waves and photons                                                                                                                                                                        |                                                           |